Prediction of drug response by using transcriptome of cancer cell lines with heterogeneity

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Introduction

Despite enormous advances in the understanding of the mechanisms underlying the malignant progression of human cancer in the last decades, targeted cancer therapy is still very challenging. The most well-known reason for poor prognosis is the high diversity of malignant tumors[1,2]. Malignant tumors are highly diverse on various levels. First, they have inter-patient tumor heterogeneity, which refers to the phenomenon that differences in tumor behavior under the same subtype and the same clinical treatment always exist between two patients[3]. On the other hand, intratumor heterogeneity indicates a difference in the tumor cell population within the same tumor specimen[2]. This heterogeneity is closely related to poor prognosis and low response rate to cancer therapy, which leads to poor survival of cancer patients[4,5]. Therefore, we have to consider tumor heterogeneity to give patients the appropriate treatment at the right time,

To overcome these issues, many have focused on patient-derived tumor xenografts (PDTXs), which retain complex intra-heterogeneity[6–8]. However, there are some limitations on using PDTXs such as high cost and animal welfare problems. Recently, a new method for isolating single-cell suspensions from PDTX tumors has been devised. Also, it is found that these PDTX-derived tumor cells (PDTCs) can reliably recapitulate the inter and intratumor genomic features observed in the original sample[9].

Herein, we tried to demonstrate that using PDTCs is better than using cell lines for predicting drug response for breast cancer. To prove this hypothesis, we compared drug response prediction models, between PDTC and patient-derived cell line data. We used four different machine learning algorithms: logistic regression, ridge regression, LASSO regression, and linear regression. We figured out that PDTC Ridge regression has a better human drug response prognosis with modest significance than any other models including Ridge regression with cell line data. I hope that our research can be helpful to the progression of patient stratification for targeted therapy (Fig. 1).

Results

Model Validation

Both the chemical properties of drugs and the biological properties of cells can affect the drug response[10]. Therefore, we combined the chemical properties of drugs and gene expression data from cell lines and PDTCs and used it as an input variable

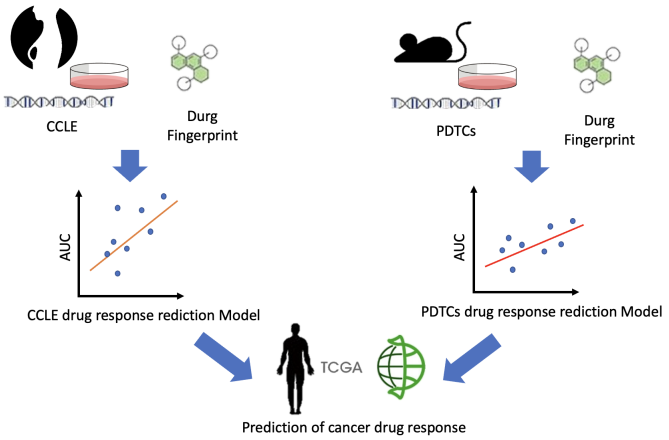

Figure 1. Schematic of the pipeline for drug response prediction task.

(shown as a schematic in Fig. 1). To represent the chemical properties of drugs in a computational way, we used the Morgan fingerprint. PDTC gene expression data consisted of more than 20,000 genes and cell line gene expression data had more than 50,000 genes. Herein, only genes in L1000 were used for circumventing the over-fitting problems and the curse of dimensionality. In total, we trained and tested the models on a dataset of 31,703 cell lines of the Cancer Cell Line Encyclopedia (CCLE) and 1,630 PDTCs, respectively.[11].

Eight models were trained in total. Four machine learning algorithms (linear, Ridge, Lasso, and logistic regression) were used to predict the area under the curve (AUC) value of given drug and PDTCs, while the same four algorithms were used to predict AUC value for the given drug and cell lines

First, we checked whether the models we made were valid or not. We separated the data into train and test datasets with the ratio of 0.8 : 0.2 and calculate the accuracy of the model trained with train data set by test data set. The result is shown in Fig. 2. Logistic regression has the best accuracy in both PDTC and cell line data. All models have an accuracy bigger than 0.5.

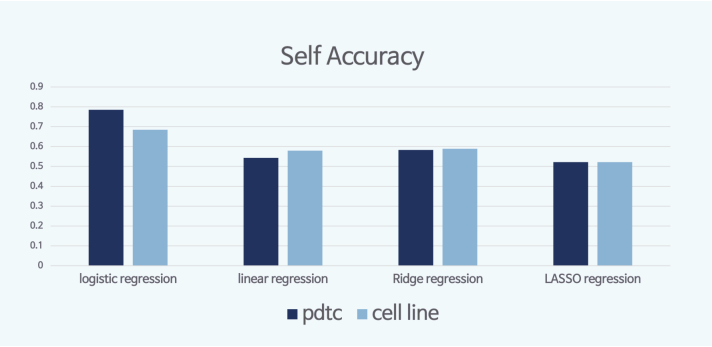

Figure 2. Accuracy of the prediction of drug response from same data type.

Human drug response prediction

The final goal of the models we made is to predict human drug responses from human gene-level expression data. Therefore, we tested the model on the dataset of 600 human gene expressions and drug response (The Cancer Genome Atlas; TCGA). As before, genes in L1000 were only used from TCGA dataset. Balanced accuracy was used as the metric to evaluate the model. We used this metric because human drug response data was highly biased to complete response. For evaluating reliability, we calculated the balanced accuracy 100 times. During the iteration, training data was sampled randomly with replacement. As a result, Ridge regression trained with PDTC data has the highest balanced accuracy considering the error from 100 iterations.

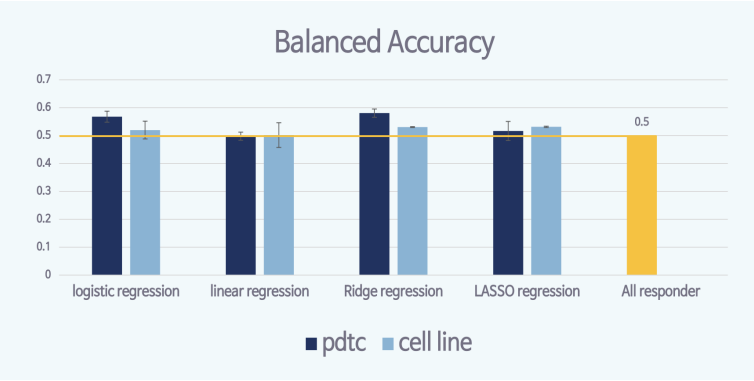

Figure 3. Balanced accuracy of the prediction using human drug response. The error bar refers to the maximum and minimum accuracy during 100 iterations.

Methods

Fingerprinting

Morgan fingerprints were first developed in 1965 for the purpose of identifying isomers[12], but later it is found that fingerprints are useful for calculating molecular similarity in large molecular databases. To represent each chemical structure of drugs in a mathematical way, we used morgan fingerprints. There are various fingerprinting algorithms. Here, we used the most popular cheminformatics package RDKit, which returns bit-vectors by default. The length of the vector was set to 2048 and the radius was set to 2.

Deep learning methods for drug response prediction

To train ML algorithms, we used expression profiles of genes of PDTCs and cell lines against drug AUC values. We used ridge, lasso, linear regression, and logistic regression, which were all implemented by scikitlearn. Default parameters were used for the ordinary least square linear regression model, ridge regression, and logistic regression. α parameter for lasso regression was 0.0001. Cross-validation was not implemented because of the time limitation.

There are several reasons why we used linear methods. First, from a practical viewpoint, linear regression and logistic regression are much simple and therefore require less expertise about machine learning. This can make the model more accessible to the bioinformatics community. Second, linear regression is interpretable. Interpretation of machine learning is an issue of importance because we cannot know biological mechanisms underlying predicted outcomes without interpretation and thus the model is less likely to reach the clinical practice level.

Conclusion

The heterogeneity of breast cancer is highly correlated to drug response and resistance. Therefore, considering heterogeneity is necessary for drug response prediction. Here, we compared the accuracy of drug response prediction models using the expression profile of PDTX-derived tumor cell (PDTC) genes and using expression profile of cell line gene to prove that prognosis is better when with PDTCs than cell lines. As a result, Ridge regression using PDTCs has the best performance. Although the range of the accuracy of the Ridge-PDTC model during 100 iterations overlapped with other models to some extent, it did not overlap with models using cell lines data. Also, considering the size difference (Cell lines have 31,703 samples while PDTCs only have 1,630 samples), the result that PDTCs have similar or slightly better accuracy than cell lines data indicates that PDTCs have better predicting performance than cell lines.

This research has several limitations. First, we did not calculate the statistical significance while model evaluation using human gene expression data. Therefore, we cannot conclude that Ridge regression using PDTCs data is significantly better than using cell lines data. Also, we did not use optimal parameters using cross-validation while using Lasso regression. Therefore, Lasso regression may have better performance with another parameter. Last, there exists a big size difference between PDTC and cell line data we used. For further study, we can calculate the exact standard deviation of accuracy from each model, find the optimal parameter for Lasso, and control the effect from size difference using a bootstrap method.

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Data Availability

1. PDTC & PDTX https://figshare.com/s/4a3f6bc543e5ba85834c
2.L1000 https://clue.io/command?q=/gene-space%20landmark
3.Cell line & Cell line drug response https://depmap.org/portal/

Code Availability

https://github.com/MunjungKim/PDTC_drug_response

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